MCQS

CS561 midterm preparation file

ZB

بسم الله الرحمن الرحيم

In the name of God , Most Gracious, Most merciful



CS561

MCQS

1. What is the primary goal of machine learning?

a) Predict future outcomes

b) Process large datasets

c) Develop accurate models

d) Automate decision-making

Answer: a

2. Which of the following is an example of supervised learning?

a) Clustering

b) Regression

c) Association rule learning

d) Reinforcement learning

Answer: b

3. In unsupervised learning, the algorithm works with:

a) Labeled data

b) Unlabeled data

c) Both labeled and unlabeled data

d) None of the above

Answer: b

4. Which algorithm is used in classification problems?

a) K-means

b) Decision trees

c) PCA

d) K-nearest neighbors

Answer: b

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5. Which of these methods can be used for dimensionality reduction?

a) Support Vector Machines

b) K-means clustering

c) Principal Component Analysis

d) Naive Bayes

Answer: c

6. Which of the following is a characteristic of deep learning?

a) Requires less data

b) Relies heavily on feature engineering

c) Uses multiple layers of computation

d) Works best with small datasets

Answer: c

7. In reinforcement learning, the agent:

a) Receives feedback in the form of rewards or penalties

b) Learns from a labeled dataset

c) Predicts outcomes based on historical data

d) None of the above

Answer: a

8. Which of the following algorithms is based on the concept of dividing the feature space into regions?

a) Support Vector Machine

b) K-means clustering

c) Decision Tree

d) Logistic Regression

Answer: c

9. Overfitting occurs when:

a) The model performs well on training data but poorly on test data

b) The model performs well on both training and test data

c) The model is too simple for the problem

d) None of the above

Answer: a

10. Which of the following is not a performance metric for classification?

a) Accuracy

b) Precision

c) Recall

d) Mean squared error

Answer: d

11. A neural network with many layers is called:

a) Shallow network

b) Deep network

c) Wide network

d) Supervised network

Answer: b

12. Which of the following is a key challenge of deep learning?

a) Lack of data

b) Overfitting

c) Feature engineering

d) Data preprocessing

Answer: b

13. In Naive Bayes classifier, the assumption is that:

a) Features are dependent on each other

b) Features are independent of each other

c) Features are linearly correlated

d) None of the above

Answer: b

14. Support Vector Machines are used for:

a) Classification

b) Regression

c) Clustering

d) Both a and b

Answer: d

15. What is the primary purpose of cross-validation?

a) To improve the speed of the model

b) To evaluate the performance of the model on unseen data

c) To reduce the complexity of the model

d) None of the above

Answer: b

16. Which of the following is true for gradient descent?

a) It converges to the global minimum in all cases

b) It is an optimization technique for minimizing a function

c) It only works for linear models

d) None of the above

Answer: b

17. Which of the following is an unsupervised learning technique?

a) K-means clustering

b) Decision trees

c) Linear regression

d) Neural networks

Answer: a

18. Which of the following is not a part of the k-nearest neighbors (KNN) algorithm?

a) Distance metric

b) Nearest neighbors

c) Voting

d) Training phase

Answer: d

19. Which of the following is an example of a continuous value prediction task?

a) Classification

b) Regression

c) Clustering

d) Association

Answer: b

20. What is the function of an activation function in a neural network?

a) To adjust the weights

b) To introduce non-linearity

c) To normalize the data

d) To reduce overfitting

Answer: b

21. Which of the following is an example of a generative model?

a) Logistic Regression

b) Naive Bayes

c) Support Vector Machines

d) K-means

Answer: b

22. In decision tree learning, the criterion used for splitting is often:

a) Gini index

b) Mean squared error

c) Entropy

d) Both a and c

Answer: d

23. Which of the following is not a form of ensemble learning?

a) Bagging

b) Boosting

c) Random Forest

d) Linear Regression

Answer: d

24. In Random Forest, the decision trees are built using:

a) Subsets of data with replacement

b) All available data

c) Clustering

d) Principal component analysis

Answer: a

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25. Which of the following is used to measure the error in regression models?

a) Accuracy

b) R-squared

c) Confusion matrix

d) F1-score

Answer: b

26. The sigmoid function outputs values in the range of:

a) 0 to 1

b) -1 to 1

c) -∞ to ∞

d) 0 to ∞

Answer: a

27. Which of the following is true for a convolutional neural network (CNN)?

a) Used primarily for sequential data

b) Contains layers like convolutional and pooling layers

c) Best suited for structured data

d) Only works with small datasets

Answer: b

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28. Which of the following is not a method for model evaluation?
a) Train-test split
b) Cross-validation
c) Grid search
d) Bootstrap
Answer: c

29. Which of the following is an advantage of deep learning over traditional machine learning?
a) Requires less data preprocessing
b) Can be easily interpreted
c) Works well with structured data
d) All of the above
Answer: a

30. In a Markov Decision Process, the agent chooses actions based on:
a) Rewards
b) States
c) Policies
d) Transitions
Answer: c

31. Which of the following is not a type of neural network?
a) Convolutional Neural Network
b) Recurrent Neural Network
c) Decision Tree
d) Multi-layer Perceptron
Answer: c

32. Which of the following is used in the training of Generative Adversarial Networks (GANs)?
a) A single network
b) Two competing networks
c) Cross-validation
d) A reinforcement learning algorithm
Answer: b

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33. In a k-means clustering algorithm, the number of clusters (k) is:

a) Chosen randomly

b) Predefined

c) Determined by the algorithm

d) Never used

Answer: b

34. Which of the following is a disadvantage of k-nearest neighbors (KNN)?

a) High memory usage

b) Overfitting

c) Slow prediction time

d) Both a and c

Answer: d

35. Which of the following models is best for modeling sequential data?

a) Decision Trees

b) Recurrent Neural Networks

c) Linear Regression

d) K-means

Answer: b

36. In principal component analysis (PCA), the components are ordered based on:

a) Variance

b) Accuracy

c) Feature importance

d) Covariance

Answer: a

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37. Which of the following is a type of hyperparameter?

a) Learning rate

b) Mean squared error

c) Training accuracy

d) Test loss

Answer: a

38. Which algorithm is commonly used for anomaly detection?

a) K-means

b) Decision Tree

c) One-class SVM

d) Naive Bayes

Answer: c

39. In the context of deep learning, what is a "dropout"?

a) A technique to prevent overfitting

b) A type of activation function

c) A regularization method for decision trees

d) None of the above

Answer: a

40. The area under the ROC curve (AUC) is used to evaluate:

a) Regression models

b) Classification models

c) Clustering models

d) None of the above

Answer: b

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41. Which of the following is not a key characteristic of reinforcement learning?

a) Agent interacts with the environment

b) The agent is rewarded or penalized

c) The agent learns from a labeled dataset

d) The agent makes decisions based on current state

Answer: c

42. The decision boundary in a linear classifier is:

a) A point

b) A line or hyperplane

c) A circle

d) A decision tree

Answer: b

43. Which of the following is an example of a generative adversarial network (GAN)?

a) Variational Autoencoder

b) Deep Convolutional GAN

c) K-means clustering

d) Support Vector Machine

Answer: b

44. What is the main difference between bagging and boosting?

a) Bagging trains multiple models independently, while boosting trains models sequentially

b) Boosting trains models independently, while bagging trains models sequentially

c) Bagging uses decision trees, while boosting uses neural networks

d) None of the above

Answer: a

45. Which of the following is true about the random forest algorithm?

a) It is an ensemble of decision trees

b) It is a form of unsupervised learning

c) It uses linear regression to build models

d) It only works with categorical data

Answer: a

46. Which of the following is a disadvantage of using a deep neural network?

a) Requires a large amount of labeled data

b) Easy to interpret

c) It's very fast during both training and prediction

d) All of the above

Answer: a

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WhatsApp:

https://whatsapp.com/channel/0029VaODheTDp2Q8YVcoON2p

Second WhatsApp channel

channel on WhatsApp:

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YouTube:

https://youtu.be/-AMj4iNz0UE?si=hg29pJZ1kujzg57J

Phone no : +92 300 5427579

Phone no: +92 330 3869586

Email: asktozb@gmail.com

Passing criteria of this course

Letter Grade	Grade Points	Equivalent Percentage
A+	4.00	90-100
A	4.00	85-89
A-	3.66 - 3.99	80-84
B+	3.33 - 3.65	75-79
B	3.00 - 3.32	71-74
B-	2.66 - 2.99	68-70
C	2.00 - 2.65	61-67
D	1.00 - 1.99	50-60
F	0.00	00-49

Passing Criteria for a Course:

The passing criteria is defined in such a way that it ensures the student shall be consistent in his studies throughout the semester.
Therefore, for passing a course/subject, student shall fulfill the following:
a) Secure minimum 20% score in Formative Assessments/Mid terms
b) Secure minimum 20% score in Final Term Examinations.
c) Secure at least 40% marks in aggregate while fulfilling the above requirements

Course Selection and Credit Hours

Q: How do I select courses when the course selection is open?
A: Follow these steps:
1. Check the credit hours allowed by your university (e.g., 21, 18, or 15 credit hours).
2. Each subject typically has 3 credit hours.
3. Divide the total allowed credit hours by 3 to determine how many subjects you can select.
Examples:
21 credit hours = 7 su18 credit hours = 6 subjects
15 credit hours = 5 subjects
Remember, the university may allow different credit hours for each student, so check your specific allowance.
By following these steps, you can make informed decisions during course selection and manage your credit hours effectively.

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Advice

Hmaesha relax ho ke parha kryn panic mat hua kryn apky parents ky bohat khawab hoty hn wo pura krny ki puri koshish kea kryn kbi b ksi pe depend na kea kryn apko bs ak insan success kr saqta ha or wo insan ap khud hn apky elawa koi nahe ap bs koshish kryn Allah pak pe strong yakeen rakha kryn or mehnat krke sb Allah ke hawaly kr dea kryn everything is possible be brave be strong stay blessed
Hum insan hamesha moat se darty or bhagty hn or moat se bachny ki koshish krty hn jabke moat ny ana hi ana ha hum

moat se nahe bach saqty humy jahanam se bachny ki koshish krni chaheay hum jahanam se bach saqty hn
Insan ko 3 chezo se dar lagta ha Moat Risk/Dolat Ezat Shohrat Fame

Moat

Humy pta hona chaheay Moat tab ani ha jab Allah chahy us se pehly puri dunya bi ak tararf ho ke apko marna chahy to apka kuch nahe beggar saqti

Risk/Dolat

Risk Dolat sb Allah pak ke hath m ha puri dunya ak taraf b ho jay na apsy apka risk cheen saqti ha na de saqti ha Ye sirf ALLAh pak ky hath m ha

Izat Zilat shohrat Fame

Izat Zilat Sirf Allah de saqty hn puri dunya mel ke b apko zra brabr b demage nahe kr saqti Agr ALLAH

Apko izat dena chahay to puri dunya ak tararf ho ke b apko 1% b nuksan nahe pohancha saqti
Phr Dar ks bat ka ??????????????
Be brave be strong Just put your trust To ALLAH
Hamesha Confident or himat se raho kbi ksi k samne mat jhuko puri dunya mel k b apka kuch b nahe beggar saqti

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MY REQUEST FROM ALL OF YOU

ZB Request from All of you My family
Mjy ap sbki help or support ki zarort ha or wo ye k hum sb Mel kr Allah ka Quran ki Urdu translation logo tak pohanchy or Quran ko samjna asaaan kryn dosro k leay sb tak Allah ka Quran pohanchy it's my compaign hum log 70+ age k ho jaty hn phr b hmy namaz tak ki translation nahe pta hoti k hum Allah pak se

Kya Dua kr rhy hn so hum youngster's ko Mel kr puri takat or energy ke sath Allah ka Quran spread krna ha
Hum puri Koshish kr rhy hn k hum sb Mel k Quran ki translation or most important topics ko maximum share kry with translation hmara maksad Quran ko spread krna ha be a part of us
m apni pocket se ye sb kr rha ho Allah ka Quran spread krny k leay apne or mene sbny Marna ha ak din to q na Allah k leay Kuch kryn apni energies apni power ko bajay negative use krny k Allah k leay invest kryn khud ko be a part of us
Agr m 23 ki age m apny sare sources use kr rha ho pocket se heavy amounts give away kr rha ho Quran spread krny k leay to ap just share to kr saqty hn itna e krdyn

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Agr ap mujsy contact na kr sky too b ap jo b hn jaha bi hn waha Allah ka quran pheelay zada se zada with urdu translation jitni himat ha utna share kryn

MY Family

ust Relax and focus! Exams are not difficult. Put in your 100% effort and trust in Allah.

AL-Quran

"And indeed, with hardship comes ease." (Quran 94:5)
Remember, I'm here to support you! Stay brave and strong!"

May Allah bless you